Aiden Sagerman

ARCHIVE GRANT ALLOCATED $6,500

One-Sentence Description

Reconstructing how topology transitioned from relying on diagram-based intuition to using algebraic logic in the 20th century.

Proposed Project

Aiden is excavating the different ways that diagrams were used in 20th-century topology research. Much of topology today is done using abstract, algebraic symbolisms, which was not the case a hundred years ago, when topology relied on geometric intuition, captured by diagrams. Aiden is interested in this historical transition from the diagram-based logic to numerical logic.

Aiden has identified three distinct phases in the history of topology:

1. Up until the late 1930s, a geometric phase
2. 1940s-1960s, an algebraic phase
3. Beginning in the 1970s, a "hybrid" phase fusing geometric and algebraic diagrams

Aiden will use the Revival funding to conduct archival visits in the United States. He's particularly interested in zooming in on the development of category theory by the topologists Samuel Eilenberg and Saunders Mac Lane from 1942 to 1945. He will reconstruct the origin of commutative diagrams, their role in the invention of category theory and the ways it subsequently changed topology.

While the project is a part of his PhD research, Aiden will produce an essay to share his research, exclusively for Analogue.

Revival Relevance

Rooted in the methods of history of science, Aiden's project reconstructs a critical transitioning moment in the history of mathematics in the mid-20th century. Aiden is driven by the question of how the underlying method of proof shifted in mathematics.

In the current rapid development of AI today, mathematics, in Aiden's words, "is in the midst of massive upheaval." Instead of fear, Aiden sees a foundational lack in how AI has been developed so far: AI may lack the creativity—lying in the training of visual reasoning—to build new theories of mathematics.

Aiden's research ambition goes beyond uncovering the events and individuals that unfolded in a short period of time. He is interested in uncovering the deeper philosophical shift that happened with how mathematicians treat reason itself, and use that to inform how we are approaching artificial intelligence today.

Wendi's Notes

1. **Aiden is particularly well-positioned to answer the research questions.**

Aiden received B.A. in Mathematics and Comparative Literature from Columbia University, M.Phil in History and Philosophy of Science from University of Cambridge, and is pursuing PhD in History of Science at Harvard

University. The combined, systematic training in mathematics and humanities makes Aiden an ideal candidate to research and write on the topic he proposed to Revival.

Aiden has presented research on several related topics at conferences:

> "Modernisms through Practices: A History of Diagrams in Algebraic Topology, 1930s-1970s," Masterclass in the Philosophy of Mathematical Practices, Centre for Logic and Philosophy of Science, Vrije Universiteit Brussel, June 2025.
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> "Algebra(s) for the 'Average Mathematician': Real Division Algebras as a Rhetorical Device in Frank Adams' Proofs of Hopf Invariant One," Luigi Maierù's International School in the History of Mathematics, May 2025.
>
> "Eugenic Internationalisms: Corrado Gini in the Interwar Americas," 12th European Spring School on History of Science and Popularization: Science, History, and Globalization, May 2025
>
> "We Have Never Been (Mathematically) Modern: Diagrammatic Practice, Algebraic Topology, and the Making of Mathematical High Modernism," Modern History of Mathematics: Looking Ahead, Isaac Newton Institute, University of Cambridge, May 2025

2. **This is an overlooked topic of research lacking in funding and support.**

Within the small field of History of Science, history of mathematics is a particularly niche topic of research. Few historians have written on the history of topology in particular. And Aiden expresses in his submission that historians often take mathematicians at face value, without thinking of them as "a tool of visual reasoning." Aiden is carrying out an ambitious work in unpacking important changes in the field through the lens of knowledge culture, approaching the history from the root question of how mathematicians do their work through the mental tools they use.

3. **Aiden's work has foundational implications for (missing) visual logic in AI development**

Aiden's focus on the history of topology, however, comes from a question with more universal implications: how did the traditions of logic and visual reasoning change in our shifting knowledge culture?

Aiden believes that a rigorous examination of visual reasoning offers a case study for the limits of AI. The current LLM models don't capture the reasoning expressed through diagrams. In Aiden's own words:

> "Examining the range of ways visual reasoning was essential to revolutionary developments in twentieth-century mathematical theory-building may provide a concrete way of understanding loose concepts such as 'creativity' that AI may struggle with. Beyond showing the limits of AI and autoformalization approaches, this project may also suggest new ways of training models that incorporate visual intuition or new ways humans can provide visual intuition for models that ultimately lead to new mathematical theories."

I am excited for Aiden to share his insights after going through a summer of archival work, which he will publish with us in the fall.